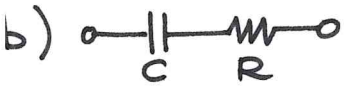
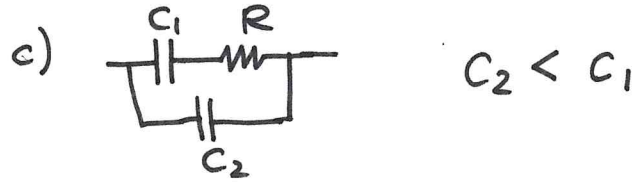
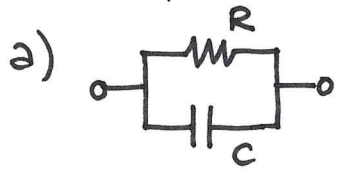


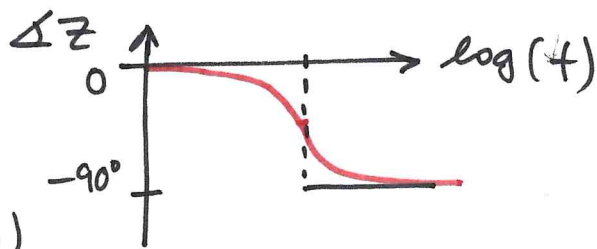
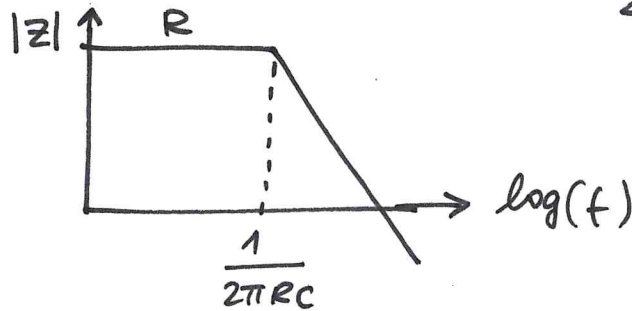
① IMPEDANCE MODELS & PLOTS

Draw the Bode plot and the Cole-Cole plot of the following equivalent impedance models:



a) $Z = R \parallel \frac{1}{sC} = \frac{R}{1 + sCR}$

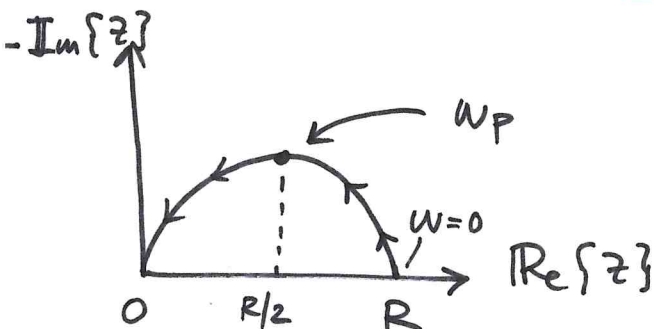
BODE:



COLE - COLE:

$$Z = \frac{R}{1 + j2\pi fCR} \cdot \frac{(1 - j\omega CR)}{(1 - j\omega CR)} = \frac{R(1 - j\omega CR)}{1 + (\omega CR)^2}$$

$$= \underbrace{\frac{R}{1 + (\omega CR)^2}}_{\text{Re}\{z\}} - j \underbrace{\frac{\omega CR^2}{1 + (\omega CR)^2}}_{\text{Im}\{z\}}$$



$\omega = 0 \Rightarrow Z = R$ (Real)

$\omega \rightarrow \infty \Rightarrow Z \rightarrow 0$ (Imaginary)

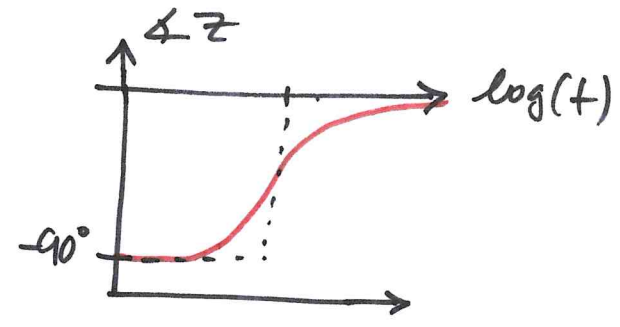
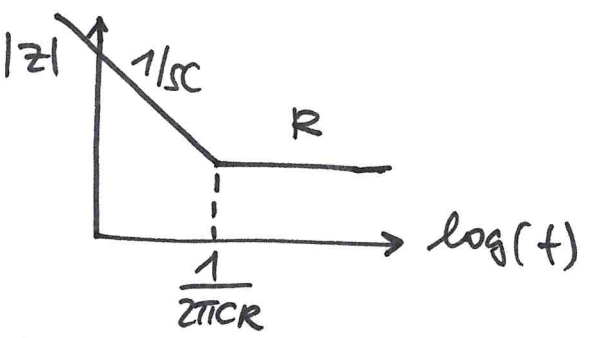
$(Z \sim \frac{1}{\omega C} \text{ capacitor})$

$\omega_P \Rightarrow \omega = \frac{1}{RC} \quad \text{Re}\{z\} = \frac{R}{1 + (\frac{1}{CR})^2} = \frac{R}{2}$

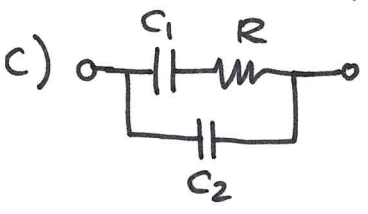
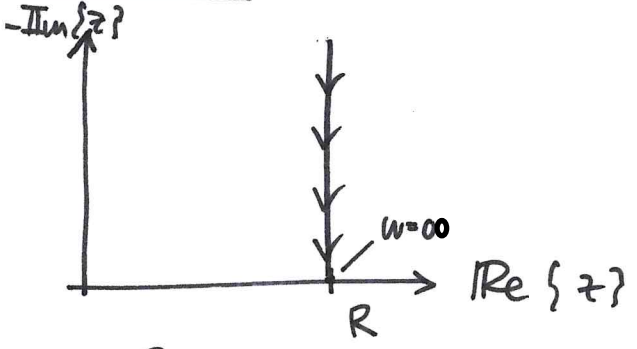
b) $Z = R + \frac{1}{sC} = R + \frac{1}{j\omega C} = R - j\frac{1}{\omega C}$

$Z = \frac{1 + sCR}{sC}$

BODE:



COLE-COLE:

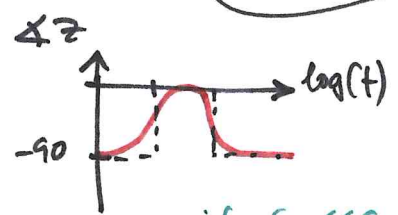
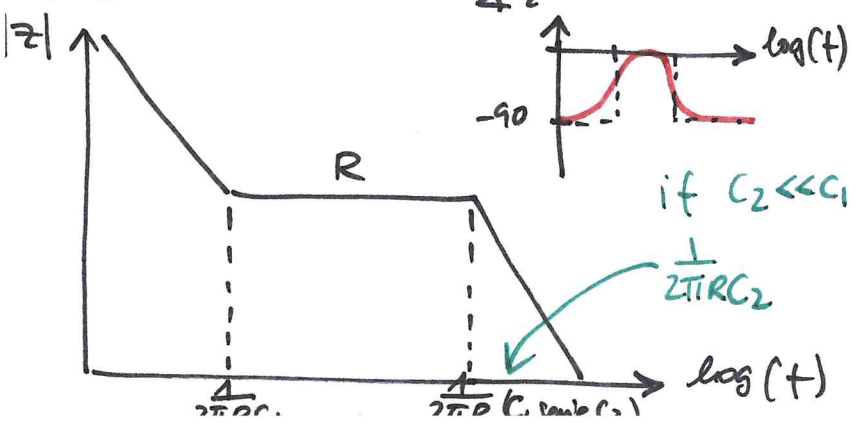


$Z = \left(R + \frac{1}{sC_1}\right) \parallel \frac{1}{sC_2} =$

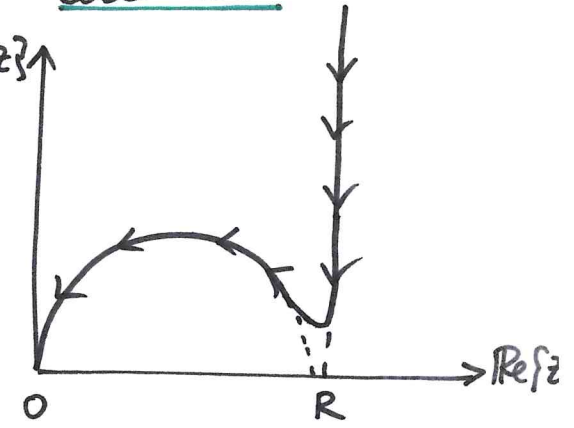
$$= \frac{1 + sC_1 R}{sC_1} \cdot \frac{1}{sC_2} = \frac{1 + sC_1 R}{\frac{1 + sRC_1}{sC_1} + \frac{1}{sC_2}} =$$

$$= \frac{1 + sC_1 R}{s(C_1 + C_2)} \cdot \frac{1}{1 + sR \frac{C_1 C_2}{C_1 + C_2}} \quad \left(\frac{C_1 \cdot C_2}{C_1 + C_2} = \text{SERIE } C_1, C_2 \right)$$

BODE:

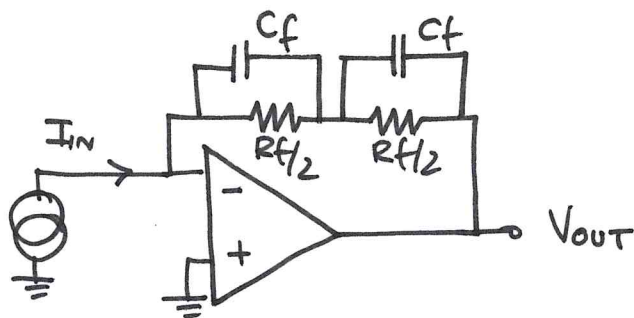


COLE-COLE:



(2) BANDWIDTH EXTENSION

Considering this topology, find the ideal transfer function and compare it with the standard transimpedance stage.



C_f = STRAY CAPACITANCE of SMT RESISTOR (0.2 pF)

$$G_{ID} = \frac{V_{OUT}}{I_{IN}} = -Z_f$$

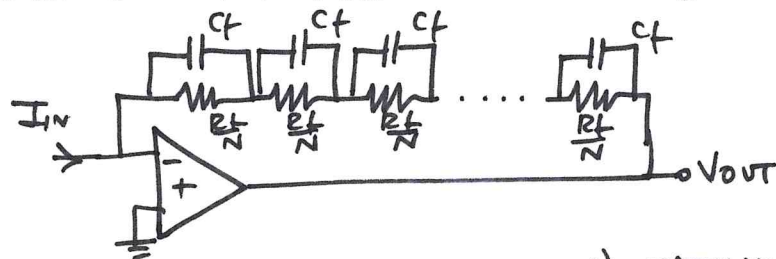
$$Z_f = \frac{R_f/2}{(1 + sC_f \cdot R_f/2)} + \frac{R_f/2}{(1 + sC_f \cdot R_f/2)}$$

$$\boxed{\frac{R_f}{1 + sC_f \cdot \frac{R_f}{2}}}$$

SAME GAIN

TWICE BANDWIDTH

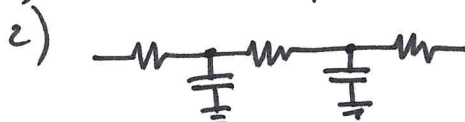
Can be extended to several resistors:



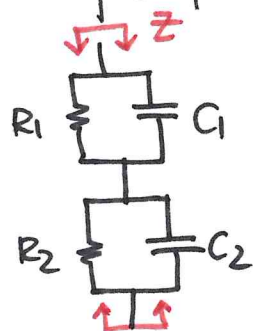
$$\frac{V_{OUT}}{I_{IN}} = - \frac{R_f}{(1 + sC_f \cdot \frac{R_f}{N})}$$

What is the limit?

1) MATCHING C_f should be equals



Many capacitors and just one pole? YES, ZERO/POLE CANCELLATION!



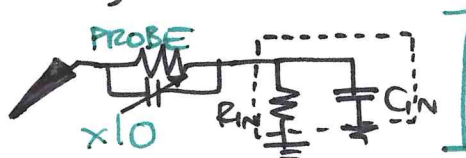
$$Z = \frac{R_1}{1 + sC_1 R_1} + \frac{R_2}{1 + sC_2 R_2} =$$

$$= \frac{R_1(1 + sC_2 R_2) + R_2(1 + sC_1 R_1)}{(1 + sC_1 R_1)(1 + sC_2 R_2)} = \frac{(R_1 + R_2) \left[1 + s \frac{R_1 R_2}{R_1 + R_2} (C_1 + C_2) \right]}{(1 + sC_1 R_1)(1 + sC_2 R_2)}$$

if $C_1 = C_2 = C$
 $R_1 = R_2 = R \Rightarrow$

$$\frac{2R \left(1 + s \frac{R}{2} \cdot 2C \right)}{(1 + sCR)(1 + sCR)} = \frac{2R}{1 + sCR} !$$

PARTITORE COMPENSATO:
OSCILLOSCOPE PROBES



(3) IMPACT OF STRAY CAP. ON STABILITY

Considering the transimpedance current amplifier, find:

a) IDEAL GAIN

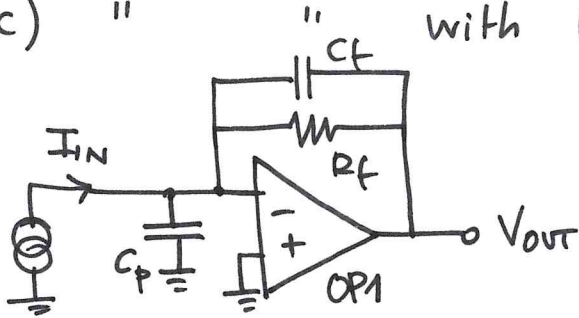
b) GBWP of OP₁ that grant stability

c) " " with R_S

$$R_f = 100 \text{ k}\Omega$$

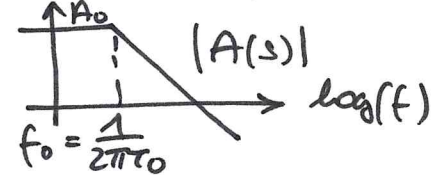
$$C_f = 1 \text{ pF}$$

$$C_p = 10 \text{ pF}$$



$$\text{OP1: } A(s) = \frac{A_0}{1 + s\tau_0}$$

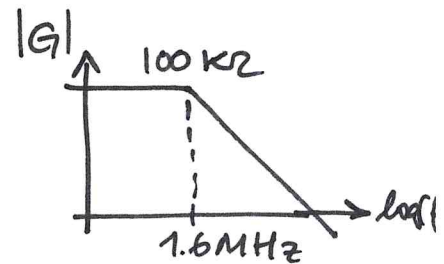
$$\text{GBWP} = A_0 \cdot f_0$$



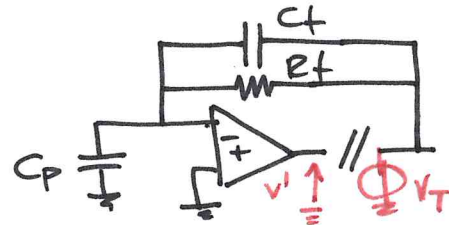
$$a) G_{ID} = \frac{V_{OUT}}{I_{IN}} = - \frac{R_f}{1 + sC_f R_f} = - \frac{100 \text{ k}\Omega}{1 + sC_f R_f}$$

$$\tau = C_f \cdot R_f = 100 \text{ ns}$$

$$f_p = \frac{1}{2\pi\tau} = 1.6 \text{ MHz}$$



b) STABILITY: → study loop gain



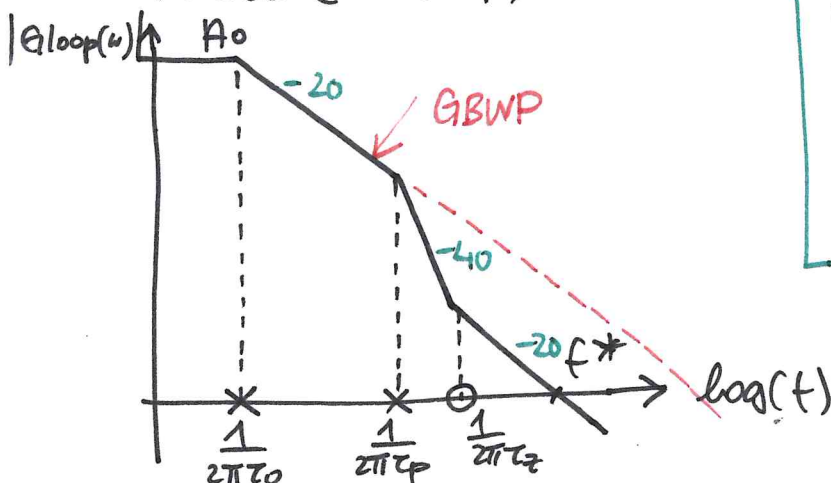
CUT the loop

$$G_{loop}(s) = \frac{V_T}{I_T} = -A(s) \cdot \frac{\frac{1}{sC_p}}{\frac{1}{sC_p} + \frac{R_f}{1 + sC_f R_f}}$$

$$= -A(s) \cdot \frac{(1 + sC_f R_f)}{1 + sR_f(C_p + C_f)} \rightarrow \text{ZERO} \quad \tau_{pole} > \tau_{ZERO}$$

POLE

$$= - \frac{A_0}{1 + s\tau_0} \frac{(1 + s\tau_z)}{(1 + s\tau_p)}$$



BODE STABILITY CRITERION:

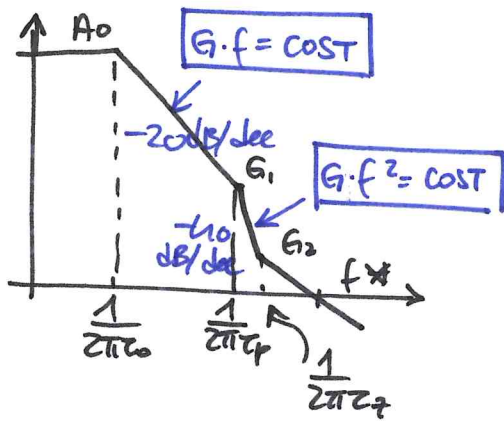
$$f^* > f_{ZERO}$$

(let's take 10 times)

$$f^* = 10 \cdot f_{ZERO} =$$

$$= 10 \cdot \frac{1}{2\pi R_f \cdot C_f}$$

Let's find the relation $GBWP \sim f^*$ in order to design a proper GBWP:



$$A_0 \cdot f_0 = GBWP = G_1 \cdot \frac{1}{2\pi R_f (C_p + C_f)}$$

$$G_1 \cdot \left(\frac{1}{2\pi R_f (C_p + C_f)} \right)^2 = G_2 \cdot \left(\frac{1}{2\pi R_f C_f} \right)^2$$

$$f^* \cdot 1 = G_2 \cdot \frac{1}{2\pi R_f C_f}$$

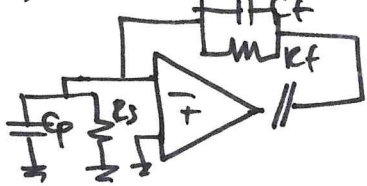
$$f^* = G_2 \cdot \frac{1}{2\pi R_f C_f} = G_1 \left(\frac{1}{2\pi R_f (C_p + C_f)} \right)^2 \cdot 2\pi R_f C_f =$$

$$= GBWP \cdot \frac{2\pi R_f C_f}{2\pi R_f (C_p + C_f)} = GBWP \cdot \frac{C_f}{C_p + C_f}$$

$$\Rightarrow GBWP = f^* \cdot \left(\frac{C_p + C_f}{C_f} \right)$$

$$GBWP > 10 \cdot \frac{1}{2\pi R_f C_f} \cdot \frac{C_p + C_f}{C_f} = 10 \cdot 1.6 \text{ MHz} \cdot \frac{11}{1} = 176 \text{ MHz}$$

c) if R_s : CLOSED-LOOP BW G_{ID} does not change



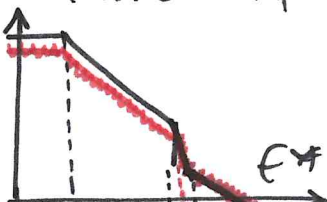
$$G_{loop}(s) = -A(s) \cdot \frac{R_s}{1 + sC_p R_s} = \frac{R_s}{\frac{R_s}{1 + sC_p R_s} + \frac{R_f}{1 + sC_f R_f}}$$

$$= -A(s) \cdot \frac{R_s}{R_s + R_f} \cdot \frac{(1 + sC_f R_f)}{1 + s(C_f + C_p) \cdot R_s // R_f}$$

$G_{loop}(0)$ is smaller τ POLE is faster (f_p higher)

THEY COMPENSATE → NO effect

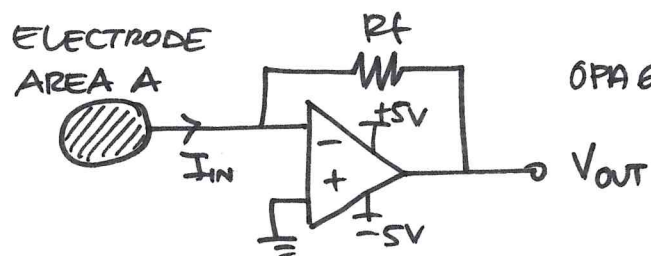
for instance: if $R_s = R_f \rightarrow -A(s) \cdot \frac{1}{2} \cdot \frac{1 + sC_f R_f}{1 + s(C_f + C_p) \cdot \frac{R_f}{2}}$



f^* is the same! i.e. at high frequency R_s is shunted by C_p

(4)

Consider the following current amplifier connected to an electrochemical sensor, of area A (radius = $25\mu\text{m}$) in PBS



OPA627 $\rightarrow$ $I_n = 2.5 \text{ fA}/\sqrt{\text{Hz}}$
 $V_m = 5 \text{ mV}/\sqrt{\text{Hz}}$

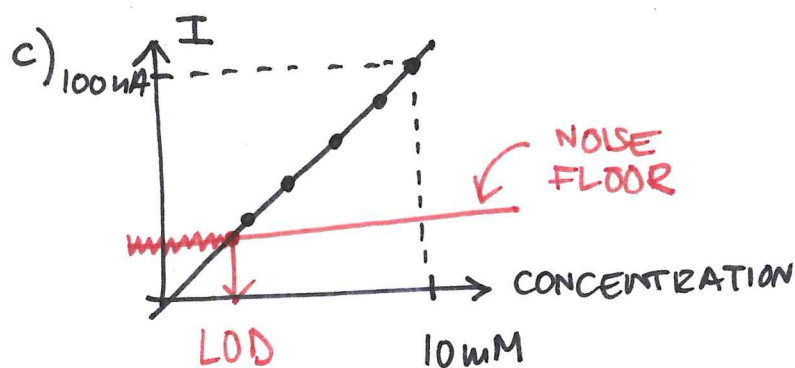
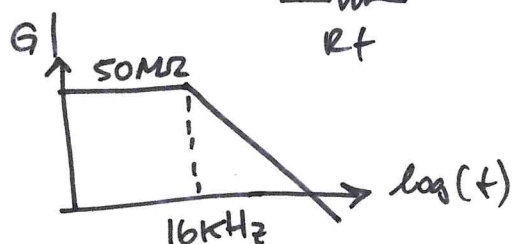
Considering an electrochemical current $I_{\text{max}} = 100 \mu\text{A}$:

- Choose the value of R_f
 - What is the amplifier bandwidth?
 - What is the LIMIT OF DETECTION?
 - If the area is reduced by a factor 10, and the scan rate increased by a factor 10, what is the new I_{max} ?
- due to ANALYTE concentration of $10 \mu\text{M}$

a) $V_{\text{OUT}} = -I_{\text{IN}} \cdot R_f$ $V_{\text{OUT}}|_{\text{max}} = (\text{if rail-to-rail}) = \pm 5\text{V}$

$R_f = \frac{5\text{V}}{I_{\text{max}}} = \frac{5\text{V}}{100 \mu\text{A}} = 50 \text{ M}\Omega$

b) C_{STRAY} $C_s = 0.2 \text{ pF}$ $\rightarrow$ $f_{\text{BW}} = \frac{1}{2\pi R_f \cdot C_f} = \frac{1}{2\pi \cdot 50 \text{ M}\Omega \cdot 0.2 \text{ pF}} \approx 16 \text{ kHz}$



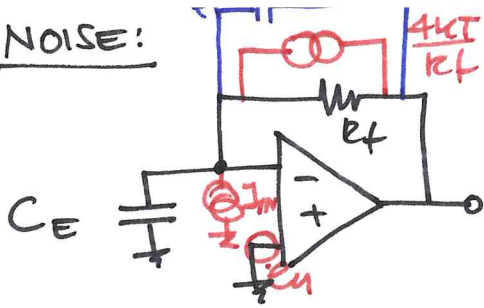
Linear dependence of analyte concentration with the current
 $\downarrow$

We need to find the noise of the amplifier.

(min resolution)

Amperometric measurement $\rightarrow$ integrate the noise spectral density over the bandwidth

NOISE:



C_E = DOUBLE LAYER CAPACITANCE

$$C_E = C_0 \cdot \text{Area}$$

C_0 = Specif capacitance, in PBS $0.1 \text{ pF}/\mu\text{m}$

$$C_E = 0.1 \frac{\text{pF}}{\mu\text{m}^2} \cdot \text{Area} = 0.1 \frac{\text{pF}}{\mu\text{m}^2} \pi \cdot r^2 = 200 \text{ pF} (196 \text{ pF}) + \cancel{0.2 \text{ pF}}^{C_f}$$

TOTAL INPUT CURRENT NOISE POWER SPECTRAL DENSITY: $S_i(f)$:

$$i_n = 2.5 \text{ fA}/\sqrt{\text{Hz}} \quad \sqrt{\frac{4kT}{R_f}} = \sqrt{4 \cdot 9 \cdot \frac{kT}{9}} \frac{1}{R_f} = 18.25 \text{ fA}/\sqrt{\text{Hz}}$$

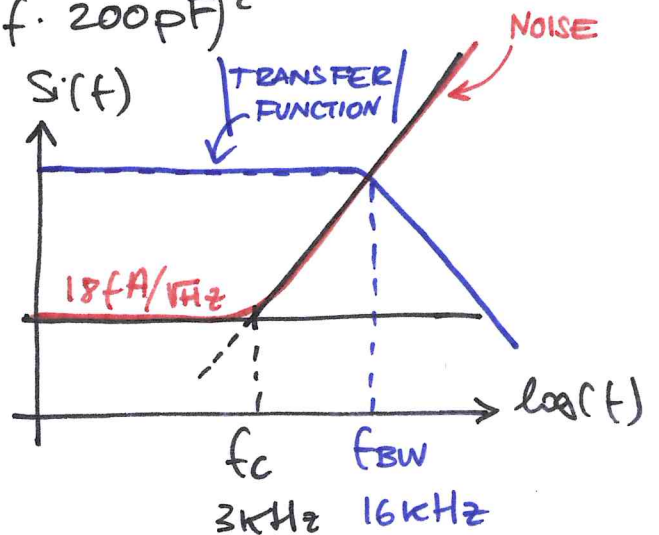
$$S_i(\text{tot}) = \sqrt{\underbrace{i_n^2}_{\text{negligible}} + \frac{4kT}{R_f} + \underbrace{(V_n \cdot 2\pi \cdot f \cdot C_E)^2}_{\text{negligible}} + \underbrace{\left(\frac{V_n}{R_f}\right)^2}_{\frac{5\text{mV}}{50\text{k}\Omega} = 0.1 \text{ fA}/\sqrt{\text{Hz}}}} =$$

$$= \sqrt{(18.2 \text{ fA}/\sqrt{\text{Hz}})^2 + \left(5 \frac{\text{mV}}{\sqrt{\text{Hz}}} \cdot 2\pi \cdot f \cdot 200 \text{ pF}\right)^2}$$

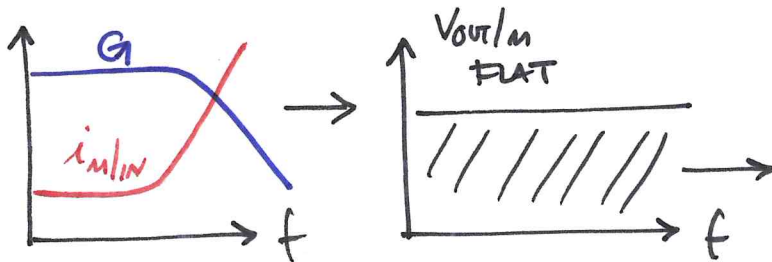
Find the corner frequency:

$$\frac{5\text{mV}}{\sqrt{\text{Hz}}} \cdot 2\pi \cdot f_c \cdot 200 \text{ pF} = 18.2 \frac{\text{fA}}{\sqrt{\text{Hz}}}$$

$$f_c = \frac{18.2 \text{ fA}}{5\text{mV} \cdot 2\pi \cdot 200 \text{ pF}} = 3 \text{ kHz}$$



If we consider only the ideal gain:



the integral is divergent!

Let's find the second pole, f^*

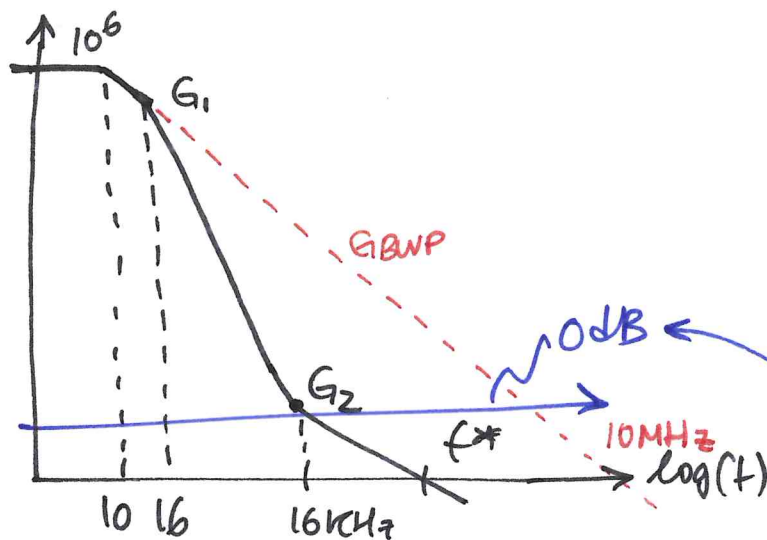
$$G_{\text{loop}}(s) = -A(s) \cdot \frac{\frac{1}{sC_E}}{\frac{1}{sC_E} + \frac{R_f}{1+sC_fR_f}} = -A(s) \cdot \frac{1+sC_fR_f}{1+s \cdot R_f(C_E+C_f)}$$

$$A(s) \Big|_{\text{OPA627}} = \frac{A_0}{1 + s\tau_0} \rightarrow 120 \text{ dB}$$

$$\text{POLE: } \frac{1}{2\pi\tau_0} = 10 \text{ kHz}$$

$$\frac{1}{2\pi R_f C_E} = 16 \text{ kHz}$$

$$\text{ZERO } \frac{1}{2\pi R_f C_f} = 16 \text{ kHz}$$



$$G_1 = \frac{10^6 \cdot 10}{16} = 625,000$$

$$G_1 \cdot 16^2 = G_2 \cdot (16 \text{ kHz})^2$$

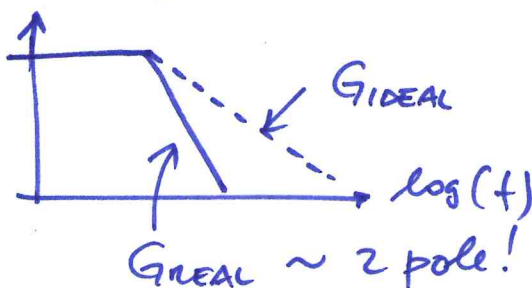
$$G_2 = G_1 \left(\frac{16 \text{ kHz}}{16 \text{ kHz}} \right)^2 = 0.625 \ll 1$$

$$G_2 \sim 1$$

AT THE LIMIT OF STABILITY!

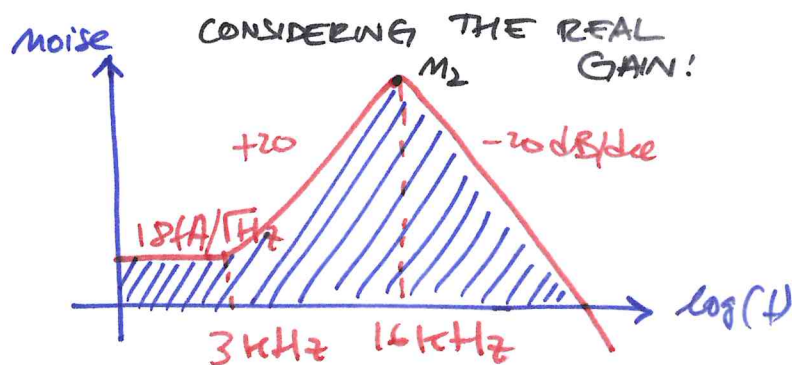
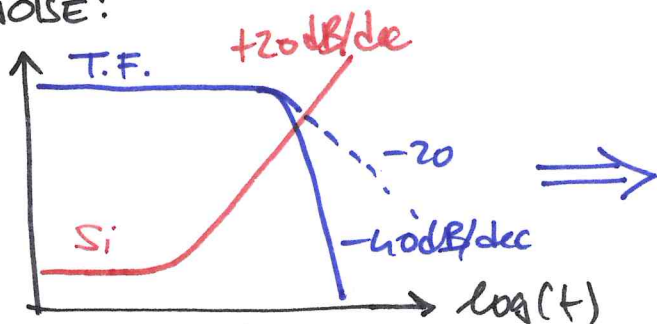
(PHASE MARGIN $\sim 45^\circ \rightarrow$ RISKY)

$$f^* \sim 16 \text{ kHz} \rightarrow$$



SOLUTION,
INCREASE C_f
(REDUCE BW)

NOISE:



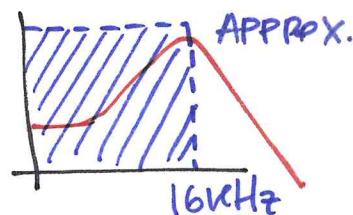
$$\frac{18 \text{ fA}/\sqrt{\text{Hz}}}{3 \text{ kHz}} = \frac{n_2}{16 \text{ kHz}}$$

$$n_2 = 18 \text{ fA}/\sqrt{\text{Hz}} \cdot \frac{16 \text{ kHz}}{3 \text{ kHz}} = 97 \text{ fA}/\sqrt{\text{Hz}}$$

$$\sigma_{\text{noise}} = \sqrt{\int_0^{f_0} S_i(f) df} \approx \sqrt{\left(\frac{97 \text{ fA}}{\sqrt{\text{Hz}}} \right)^2 \cdot 16 \text{ kHz}} = 12.3 \text{ pA}_{\text{rms}}$$

APPROX. RUDE

(SHOULD USE EQUIVALENT NOISE BW = $\frac{\pi}{2} \cdot f_{-3\text{dB}}$)



$$LOD = \frac{12.3 \text{ pA}_{rms}}{100 \text{ nA}} \cdot 10 \text{ mM} = 1.2 \mu\text{M}$$

d) $A' = \frac{A}{10} \Rightarrow I' = \frac{I}{10}$ (macroscopic planar diffusion regime)

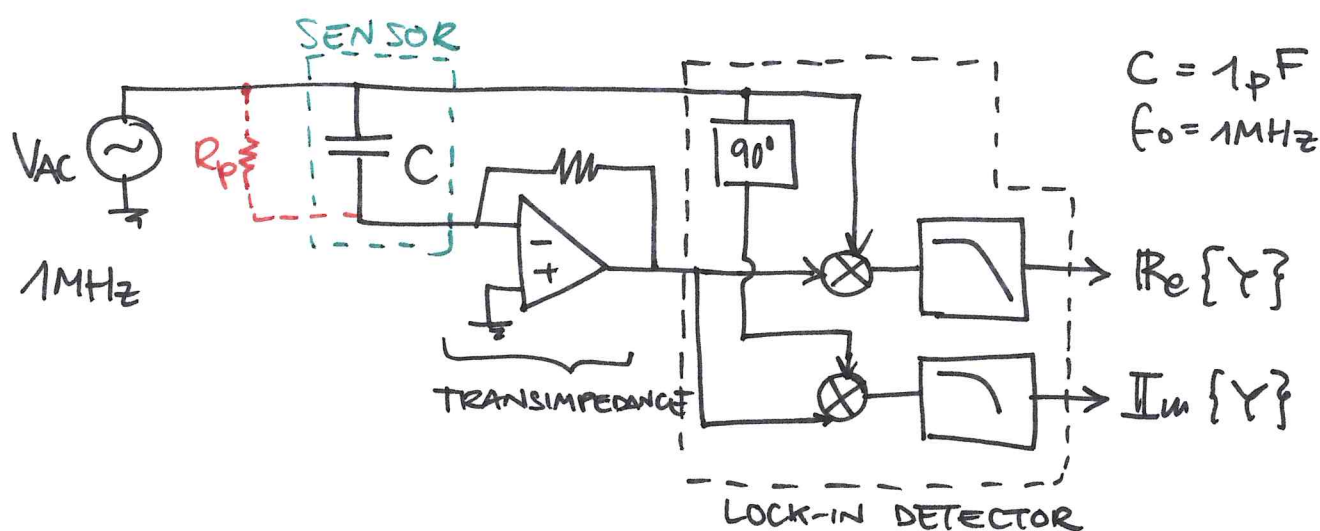
$$SR' = SR \cdot 10 \Rightarrow I' = I \cdot \sqrt{10} = I \cdot 3.1$$

$$I_{\max}' = 100 \mu\text{A} \cdot \frac{1}{10} \cdot \sqrt{10} = \boxed{31.6 \text{ nA}}$$

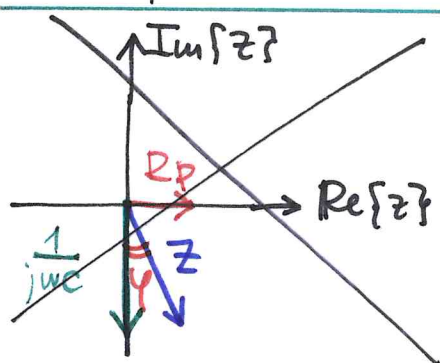
$$I_p \propto A \cdot \sqrt{SR}$$

(5)

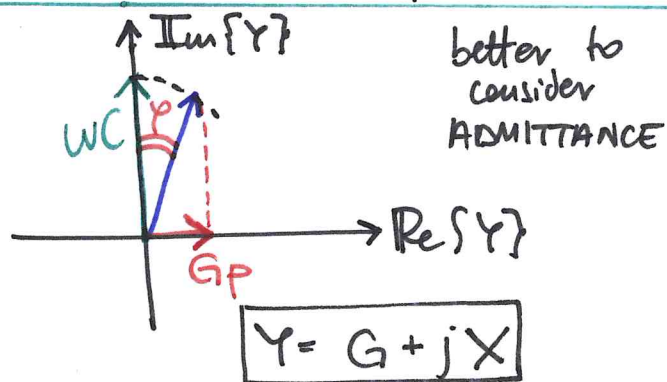
Consider the following circuit for CAPACITIVE sensing:



Find the minimum bandwidth of the transimpedance amplifier that provides an apparent parallel resistance $R_p > 16M\Omega$



$$Y = j\omega C$$



$$G_p = \frac{1}{R_p} = \frac{1}{16M\Omega} = 62.5 \cdot 10^{-9} [S] \text{ max. real admittance apparent}$$

$$|Y| = |j\omega C| = 2\pi f \cdot C = 2\pi \cdot 1MHz \cdot 1pF = 6.28 \cdot 10^{-6} S$$

$$G_p = |Y| \cdot \sin \varphi \rightarrow \varphi = \arcsin\left(\frac{G_p}{|Y|}\right) = 0.57^\circ \text{ MAX. TOLERABLE PHASE SHIFT!}$$

$$H_T = \frac{R_T}{1 + s\tau} \text{ What BW gives } 0.57^\circ \text{ at } 1MHz?$$

$$\Delta H_T(1MHz) = 0.57^\circ \rightarrow \arctan(f/f_p) = \arctan\left(\frac{1MHz}{f_p}\right) = 0.57^\circ$$

$$f_p = \frac{1MHz}{\tan(0.57^\circ)} = \boxed{100MHz}$$